

Overview of Stable Diffusion

Subject: Stable Diffusion

$$\mathcal{L} = \mathbb{E}_{x, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

1.

Getty Images v. Stability AI In January 2023, Getty Images initiated legal proceedings against Stability AI in the English High Court, alleging significant infringement of its intellectual property rights. Getty Images claims that Stability AI "scraped" millions of images from Getty's websites without consent and used these images to train and develop its deep-learning Stable Diffusion model. Key points of the lawsuit included:

2.

Capabilities The Stable Diffusion model supports the ability to generate new images from scratch through the use of a text prompt describing elements to be included or omitted from the output. Existing images can be re-drawn by the model to incorporate new elements described by a text prompt (a process known as "guided image synthesis") through its diffusion-denoising mechanism. In addition, the model also allows the use of prompts to partially alter existing images via inpainting and outpainting, when used with an appropriate user interface that supports such features, of which numerous different open source implementations exist. Stable Diffusion is recommended to be run with 10 GB or more VRAM, however users with less VRAM may opt to load the weights in float16 precision instead of the default float32 to tradeoff model performance with lower VRAM usage.

3.

Training procedures The model was initially trained on the laion2B-en and laion-high-resolution subsets, with the last few rounds of training done on LAION-Aesthetics v2 5+, a subset of 600 million captioned images which the LAION-Aesthetics Predictor V2 predicted that humans would, on average, give a score of at least 5 out of 10 when asked to rate how much they liked them. The LAION-Aesthetics v2 5+ subset also excluded low-resolution images and images which LAION-5B-WatermarkDetection identified as carrying a watermark with greater than 80% probability. Final rounds of training additionally dropped 10% of text conditioning to improve Classifier-Free Diffusion Guidance. The model was trained using 256 Nvidia A100 GPUs on Amazon Web Services for a total of 150,000 GPU-hours, at a cost of \$600,000.

4.

Development Stable Diffusion originated from a project called Latent Diffusion, developed in Germany by researchers at LMU Munich in Munich and Heidelberg University. Four of the original 5 authors (Robin Rombach, Andreas Blattmann, Patrick Esser and Dominik Lorenz) later joined Stability AI and released subsequent versions of Stable Diffusion. The technical license for the model was released by the CompVis group at LMU Munich. Development was led by Patrick Esser of Runway

and Robin Rombach of CompVis, who were among the researchers who had earlier invented the latent diffusion model architecture used by Stable Diffusion. Stability AI also credited EleutherAI and LAION (a German nonprofit which assembled the dataset on which Stable Diffusion was trained) as supporters of the project.

5.

License Unlike models like DALL-E, Stable Diffusion makes its source code available, along with the model (pretrained weights). Prior to Stable Diffusion 3, it applied the Creative ML OpenRAIL-M license, a form of Responsible AI License (RAIL), to the model (M). The license prohibits certain use cases, including crime, libel, harassment, doxing, "exploiting ... minors", giving medical advice, automatically creating legal obligations, producing legal evidence, and "discriminating against or harming individuals or groups based on ... social behavior or ... personal or personality characteristics ... [or] legally protected characteristics or categories". The user owns the rights to their generated output images, and is free to use them commercially. Stable Diffusion 3.5 applies the permissive Stability AI Community License while commercial enterprises with revenue exceed \$1 million need the Stability AI Enterprise License. As with the OpenRAIL-M license, the user retains the rights to their generated output images and is free to use them commercially.

6.

An "embedding" can be trained from a collection of user-provided images, and allows the model to generate visually similar images whenever the name of the embedding is used within a generation prompt. Embeddings are based on the "textual inversion" concept developed by researchers from Tel Aviv University in 2022 with support from Nvidia, where vector representations for specific tokens used by the model's text encoder are linked to new pseudo-words. Embeddings can be used to reduce biases within the original model, or mimic visual styles. A "hypernetwork" is a small pretrained neural network that is applied to various points within a larger neural network, and refers to the technique created by NovelAI developer Kurumuz in 2021, originally intended for text-generation transformer models. Hypernetworks steer results towards a particular direction, allowing Stable Diffusion-based models to imitate the art style of specific artists, even if the artist is not recognised by the original model; they process the image by finding key areas of importance such as hair and eyes, and then patch these areas in secondary latent space. DreamBooth is a deep learning generation model developed by researchers from Google Research and Boston University in 2022 which can fine-tune the model to generate precise, personalised outputs that depict a specific subject, following training via a set of images which depict the subject.

7.

Stable Diffusion Demo "Step by Step visual introduction to Diffusion Models. - Blog by Kemal Erdem". Retrieved August 31, 2024. "U-Net for Stable Diffusion". U-Net for Stable Diffusion. Retrieved August 31, 2024. Interactive Explanation of Stable Diffusion "We Are All Raw Material for AI": Investigation on sensitive and private data in Stable Diffusions training data

8.

Getty Images asserted that the training and development of Stable Diffusion involved the unauthorized use of its images, which were downloaded on servers and computers that were potentially in the UK. However, Stability AI argued that all training and development took place outside the UK, specifically

in U.S. data centers operated by Amazon Web Services. Stability AI applied for reverse summary judgment and/or strike out of two claims: the training and development claim, and the secondary infringement of copyright claim. The High Court, however, refused to strike out these claims, allowing them to proceed to trial. The court is to determine whether the training and development of Stable Diffusion occurred in the UK, which is crucial for establishing jurisdiction under the UK's Copyright, Designs and Patents Act 1988 (CDPA). The secondary infringement claim revolved around whether the pre-trained Stable Diffusion software, made available in the UK through platforms like GitHub, HuggingFace, and DreamStudio, constitutes an "article" under sections 22 and 23 of the CDPA. As of November 4, 2025 Getty Images largely lost the lawsuit over AI image generator to Stability AI, prompting Getty and some lawyers to call for stronger protections for copyright owners in Britain.